

3.5 Control Structures Overview

کنٹرول اسٹرکچرز (Control Structures) طے کرتے ہیں کہ پروگرام کی ہدایات کس ترتیب سے چلیں گی۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 08 of 13

★ Golden

3.5 Control Structures Overview ★ Golden

LEARN IT

Control structures decide the **order in which statements are executed** in a program (the "flow of control"). Almost every program is built from just three basic structures:

1. **Sequence** — do the steps one after another, top to bottom.
2. **Selection** — choose a path depending on a condition (`if` , `else`).
3. **Repetition** — repeat steps (loops: `for` , `while`).

WORKED EXAMPLE

Morning routine as control structures:

- Sequence: wake up → brush teeth → have breakfast.
- Selection: **if** it is raining, take an umbrella; **else** take a cap.
- Repetition: **while** the bus has not come, keep waiting.

REMEMBER IT

Remember **"S-S-R"**: **S**equence, **S**election, **R**epetition — "Step, Select, Repeat".

COMMON MISTAKE

Selection and repetition both use conditions, but they are different: selection runs a block **at most once**; repetition can run a block **many times**.

اردو میں سمجھیں

کنٹرول اسٹرکچرز (Control Structures) طے کرتے ہیں کہ پروگرام کی ہدایات کس ترتیب سے چلیں گی۔ تین بنیادی اسٹرکچر ہیں: سیکوئنس (ترتیب وار)، سلیکشن (شرط کی بنیاد پر انتخاب) اور ریپیٹیشن (دہرانا یعنی لوپ)۔

EXAM TIP

"Name and explain the three basic control structures" is a very common short question. Give a one-line definition and a two-line Python example for each.

CHECK YOURSELF

1. Name the three basic control structures.
2. Which structure is used for "keep asking for the password until it is correct"?

Answers: 1. Sequence, selection, repetition. 2. Repetition (a `while` loop).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The three basic control structures are:

- a. Sequence, selection, repetition
- b. AND, OR, NOT
- c. OSI 1,2,3
- d. Mean, median, mode

MCQ 2. Sequence means:

- a. One after another
- b. Skip around
- c. Loop forever
- d. Sort first

MCQ 3. Selection uses:

- a. if / elif / else
- b. only for
- c. only def
- d. only import

MCQ 4. Repetition uses:

- a. for / while
- b. print only
- c. class only
- d. try only

MCQ 5. Almost every useful program mixes:

- a. All three
- b. Only comments
- c. Only gates
- d. Only charts

MCQ 6. A flowchart diamond is:

- a. A decision (selection)
- b. A process always
- c. A file
- d. A stack

Extra short questions

1. Define the three control structures.
2. Give a school example of each.
3. What happens if you only have sequence?
4. How do you show them in a flowchart?

Extra long question

1. The shopping-bill program uses all three structures. Identify one line of each and explain why the bill would be wrong if that structure vanished.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Sequence: order. Selection: choose a path. Repetition: repeat a block.
2. Seq: enter marks then print. Sel: if fail, reprint. Rep: for each student.
3. The program cannot react or repeat — it is a straight recipe.
4. Rectangles process, diamonds decisions, arrows loops back.

Long

1. Seq: print header then loop. Sel: if qty invalid. Rep: while more items. Without loop only one item; without if bad qty; without sequence output is unordered.